Design Fundamentals 2/4

Topics 7–12

7. Caching

- Client, server, CDN, DB caches
- Write-through vs write-back
- Eviction policies LRU/LFU
- Staleness and invalidation

8. Proxies

- Forward proxy (acts for the client) vs reverse proxy (acts for the server: LB, TLS termination, caching)
- Nginx

9. Load balancers

- Round robin, weighted, least-connections, IP/consistent hashing
- Health checks
- L4 vs L7

10. Hashing

- Consistent hashing and rendezvous hashing keep remapping small when servers join or leave

11. Relational databases

- SQL, ACID transactions, indexes (B-trees), strong consistency, joins

12. Key-value stores

- Simple get/put, very fast, flexible
- Redis (memory), DynamoDB, etcd/ZooKeeper (strongly consistent config)

Recap

- 7. Caching
- 8. Proxies
- 9. Load balancers
- 10. Hashing
- 11. Relational databases
- 12. Key-value stores